



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Retail Inventory Movement Tracking and Reporting System using SAP

The domain of the Project:
SAP ABAP APPLICATION

Mentors (and their designation):

Mr. Santosh Pal
SURE Trust Trainer

Mr. Avishek Kumar Chourasiya

Period of the project

May 2026 to July 2026



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Declaration

The project titled “***Retail Inventory Movement Tracking and Reporting System using SAP***” has been mentored by **Mr. Santosh Pal**, organised by SURE Trust, from **May 2026 to July 2026**, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Mr. Avishek Kumar Chourasiya

Mr. Santosh Pal

SURE Trust Trainer

SURE Trust (SURE ProEd)

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Overview of the Project Objectives

The Retail Inventory Movement Tracking and Reporting System using SAP was developed to demonstrate how SAP solutions can improve inventory management and supply chain operations in the retail sector. The primary objective of the project was to provide efficient tracking of inventory movements, enable detailed reporting of material transactions, and support better business decision-making through automated and centralized reporting mechanisms.

Methods Used

- Utilized SAP ABAP for report development and data processing.
- Retrieved inventory movement data from SAP tables MKPF and MSEG.
- Developed Interactive ALV Reports for summary and detailed transaction analysis.
- Implemented drill-down functionality for detailed inventory movement tracking.
- Generated professional reports using SAP Smart Forms.
- Integrated Excel export functionality for data analysis and reporting.

Key Findings

- Improved visibility of inventory movement and stock transactions.
- Reduced manual effort involved in report preparation and analysis.
- Enabled faster access to detailed inventory information through Interactive ALV.
- Enhanced reporting accuracy by retrieving real-time SAP data.
- Demonstrated the effectiveness of SAP in supporting retail supply chain operations.



Recommendations

Retail organizations should adopt SAP-based inventory monitoring and reporting solutions to improve supply chain visibility, enhance reporting accuracy, and streamline inventory management processes. Automated reporting tools can help businesses make faster and more informed decisions while reducing operational inefficiencies.

Highlights for Executives and Stakeholders

The project demonstrates the practical application of SAP in improving retail inventory management and supply chain operations. By leveraging SAP ABAP technologies, the solution provides a centralized platform for monitoring inventory movements, generating business reports, and supporting operational decision-making. The implementation highlights how SAP can transform traditional inventory tracking processes into a more efficient, transparent, and data-driven system.

Key highlights include:

- Real-time visibility into inventory and material movement transactions.
- Improved accuracy and reliability of inventory reporting.
- Faster access to business-critical information through Interactive ALV reports.
- Reduced dependency on manual data collection and report preparation.
- Enhanced inventory monitoring across warehouses and retail locations.
- Efficient tracking of goods receipts, goods issues, and stock transfers.
- Improved operational transparency through detailed transaction-level reporting.
- Support for informed decision-making using accurate and up-to-date business data.
- Automated generation of printable reports using SAP Smart Forms.
- Easy export of data to Excel for analysis, auditing, and management review.
- Better control over inventory-related business processes



Background and Context of the Project

The retail industry relies heavily on efficient inventory management and supply chain operations to ensure that products are available to customers at the right time and place. With the increasing volume of transactions and inventory movements, manual tracking methods often lead to delays, inaccuracies, and operational inefficiencies. SAP provides integrated solutions that help retail organizations manage inventory, monitor stock movements, and generate business reports in real time. This project focuses on demonstrating how SAP can be utilized to improve inventory visibility and streamline retail supply chain operations.

Problem Statement or Goals of the Project

Retail organizations frequently face challenges in tracking inventory movements, maintaining accurate stock records, and generating timely reports for decision-making. Manual processes can result in data inconsistencies and increased operational effort. The goal of this project is to develop a SAP-based reporting solution that enables efficient monitoring of inventory transactions, provides detailed visibility into material movements, and supports informed business decisions through automated reporting and analysis.

Scope and Limitations of the Project

Scope

- Monitoring inventory and material movement transactions.
- Generating Interactive ALV reports for inventory analysis.
- Displaying detailed transaction information through drill-down functionality.



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- Creating printable reports using SAP SmartForms.
- Exporting inventory data to Excel for further analysis.

Limitations

- The project focuses primarily on inventory movement reporting and analysis.
- Advanced retail functionalities such as sales forecasting and demand planning are not included.
- The system operates on available SAP transactional data and does not integrate with external applications.
- The project is developed as a learning and demonstration solution rather than a full-scale enterprise implementation.

Innovation Component in the Project

The innovative aspect of the project lies in the integration of ***Interactive ALV reporting***, ***Smart Form generation***, and ***Excel export functionality*** within a single SAP-based solution.

The system provides users with real-time access to inventory movement information and enables detailed transaction analysis through an interactive interface. By automating reporting processes and improving data accessibility, the project demonstrates how SAP technology can enhance operational efficiency, transparency, and decision-making in retail supply chain management.



Project Objectives

Project Objectives & goals

- To develop a SAP-based inventory movement tracking and reporting system for retail supply chain operations.
- To retrieve and analyse inventory transaction data using SAP tables MKPF and MSEG.
- To implement Interactive ALV reports for efficient monitoring and analysis of inventory movements.
- To provide drill-down functionality for detailed transaction-level information.
- To generate professional inventory reports using SAP Smart Forms.
- To enable Excel export functionality for reporting and business analysis purposes.
- To improve inventory visibility and reporting accuracy within retail operations.
- To reduce manual effort involved in inventory tracking and report generation.
- To demonstrate the practical application of SAP technologies in retail supply chain management.
- To gain hands-on experience in SAP ABAP development, reporting, and business process automation.

Goals

The goal of the Retail Inventory Movement Tracking and Reporting System using SAP is to demonstrate how SAP can be leveraged to improve inventory management, reporting efficiency, and supply chain visibility within the retail sector. The project aims to provide a centralized and automated reporting solution that enables organizations to monitor inventory movements, access accurate business information, and support effective decision-making. By integrating Interactive ALV reporting, SmartForms, and inventory transaction analysis, the project showcases how SAP helps retail businesses optimize operations, enhance productivity, and maintain better control over inventory-related processes.



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Expected Outcomes and Deliverables

The project is expected to provide an efficient SAP-based reporting solution that enables organizations to monitor inventory movements, generate accurate reports, and improve supply chain visibility. The system will support operational decision-making by providing real-time access to inventory data and reducing dependency on manual reporting processes.

Expected Outcomes:

- Improved inventory tracking and monitoring capabilities.
- Enhanced reporting accuracy and operational efficiency.
- Faster access to inventory-related information.
- Better visibility into retail supply chain activities.
- Improved support for business and operational decision-making.

Project Deliverables:

- Interactive ALV Report.
- SmartForm-Based Report.
- Excel Export Functionality.
- SAP ABAP Program Source Code.
- Project Documentation.
- System Screenshots and Output Reports.



Methodology / Technology Used

The **Retail Inventory Movement Tracking and Reporting System using SAP** was developed using SAP ABAP technologies to monitor and analyze inventory movement transactions within retail supply chain operations. The system retrieves inventory-related information from SAP database tables and presents it through interactive reporting mechanisms. The methodology focused on collecting material movement data, processing transaction records, generating reports, and providing users with detailed visibility into inventory activities.

Technologies Used:

- SAP ABAP
- Interactive ALV Reports
- SAP Smart Forms
- SAP Data Dictionary
- Internal Tables and Work Areas
- Event Handling and Drill-Down Functionality

Tools / Software Used

The project was developed using various SAP development tools and reporting technologies to support inventory tracking, report generation, and data analysis within the retail supply chain environment.

SAP Development Tools

- SAP GUI (Graphical User Interface)
- ABAP Workbench (SE38)
- ABAP Editor



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- Smart Forms
- Function Builder (SE37)
- Object Navigator (SE80)
- SAP Menu and Navigation Tools

SAP Technologies

- SAP ABAP Programming
- Interactive ALV Reports
- Classical ALV Reports
- SmartForms
- Internal Tables
- Work Areas
- Function Modules
- Open SQL
- Events and Interactive Programming

SAP Database Tables

- MKPF (Material Document Header)
- MSEG (Material Document Segment)
- MARA (Material Master)
- MAKT (Material Descriptions)
- T001W (Plant Information)
- T001L (Storage Location Information)

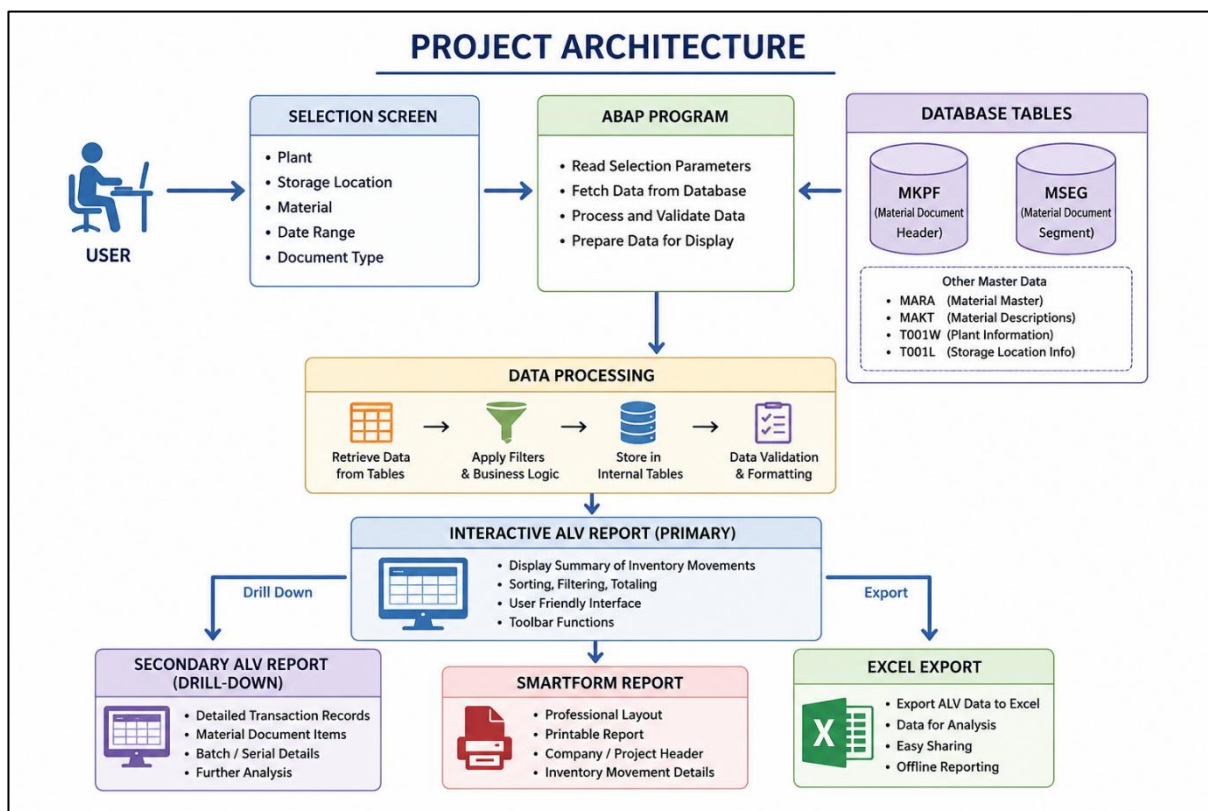


Data Collection Approach

The project utilizes transaction and material movement data stored within SAP ERP tables. Inventory movement information is retrieved from MKPF and MSEG tables, which contain material document headers and item-level transaction details respectively. The collected data is processed and displayed through Interactive ALV reports, allowing users to analyze inventory activities efficiently.

Project Architecture

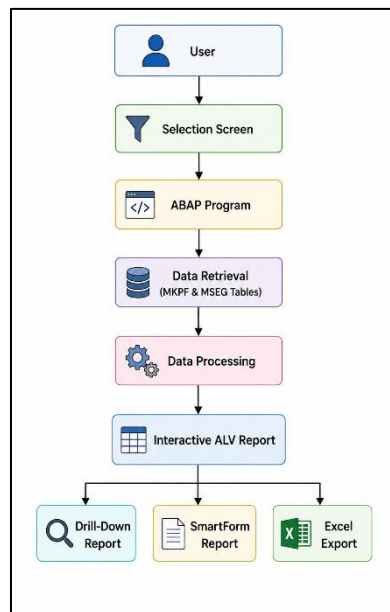
The system follows a structured reporting architecture where inventory transaction data is retrieved from SAP database tables and processed through ABAP programs before being presented to end users.





Architecture Flow:

User → Selection Screen → ABAP Program → MKPF & MSEG Tables → Internal Tables →
Interactive ALV Report → Smart Form Report / Excel Export



: Workflow

Architecture Description:

- The user enters selection criteria through the SAP selection screen.
- The ABAP program retrieves relevant inventory movement data from MKPF and MSEG tables.
- Retrieved data is processed using internal tables and business logic.
- Interactive ALV displays summarized inventory information.
- Drill-down functionality provides detailed transaction records.
- SmartForms generate printable reports.
- Excel Export functionality enables further analysis and reporting.



Final Project Working Screenshots and Results

The project successfully generated Interactive ALV reports displaying inventory movement information retrieved from SAP tables. Users can navigate from summary-level reports to detailed transaction records through drill-down functionality. SmartForms provide printable reports, while Excel Export allows users to download and analyze inventory data outside SAP.

The screenshot shows a SAP 'alv report' selection screen. At the top, the text 'alv report' is displayed. Below it, there is a green checkmark icon. The main area of the screen contains a search field with the text 'SO_MBLNR' and a blue rectangular selection box. To the right of the search field, the word 'to' is followed by another empty search field. On the far right, there is a small icon of a document with a yellow arrow pointing to it.

Figure 1: Report Selection Screen

The selection screen enables users to specify material document numbers for retrieving inventory movement data. The entered criteria are processed by the SAP ABAP program to generate the Interactive ALV report.



alv report			
			
Material Doc	Year	Doc Date	Posting Date
0000000047	2024	14.01.2024	14.01.2024
0000000048	2024	14.01.2024	14.01.2024
0000000049	2024	14.01.2024	14.01.2024
0000000050	2024	16.01.2024	16.01.2024
0000000051	2024	16.01.2024	16.01.2024
0000000052	2024	16.01.2024	16.01.2024
0000000053	2024	16.01.2024	16.01.2024
0000000054	2024	16.01.2024	16.01.2024
0000000055	2024	16.01.2024	16.01.2024
0000000056	2024	16.01.2024	16.01.2024
0000000057	2024	16.01.2024	16.01.2024
0000000058	2024	16.01.2024	16.01.2024
0000000059	2024	16.01.2024	16.01.2024
0000000060	2024	16.01.2024	16.01.2024
0000000061	2024	16.01.2024	16.01.2024
0000000062	2024	16.01.2024	16.01.2024
0000000063	2024	16.01.2024	16.01.2024
0000000065	2024	19.01.2024	19.01.2024
0000000066	2024	19.01.2024	19.01.2024
0000000071	2024	03.02.2024	03.02.2024
0000000072	2024	03.02.2024	03.02.2024
0000000081	2024	06.02.2024	06.02.2024
0000000091	2024	15.03.2024	15.03.2024
0000000092	2024	15.03.2024	15.03.2024
0000000093	2024	15.03.2024	15.03.2024
0000000095	2024	17.03.2024	17.03.2024
0000000096	2024	17.03.2024	17.03.2024
0000000101	2024	21.03.2024	21.03.2024
0000000102	2024	21.03.2024	21.03.2024
0000000103	2024	21.03.2024	21.03.2024
0000000104	2024	21.03.2024	21.03.2024
0000000105	2024	21.03.2024	21.03.2024
0000000106	2024	21.03.2024	21.03.2024

Figure 2: Primary Interactive ALV Report

This screen displays the primary Interactive ALV report containing material document details retrieved from SAP tables MKPF and MSEG. It provides key information such as Material Document Number, Fiscal Year, Document Date, and Posting Date. The report enables users to view and analyze inventory movement transactions in a structured and user-friendly format.

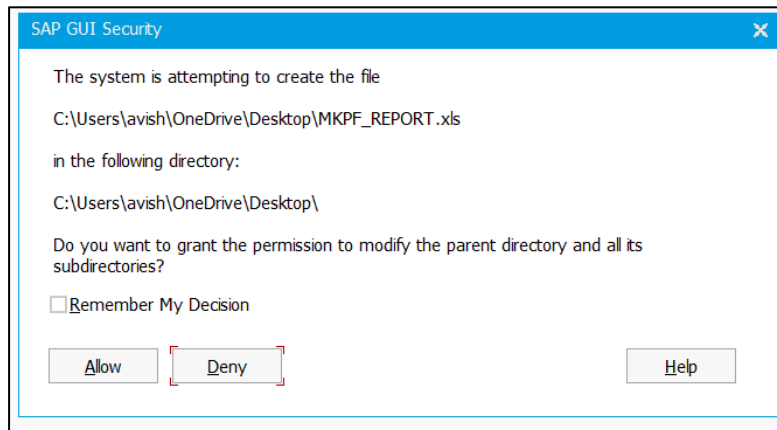


Figure 3: Excel Export Permission Window

This screen displays the SAP GUI security prompt that appears when the system attempts to create and save the Excel report file on the local machine. The user must grant permission to allow the export process to proceed .

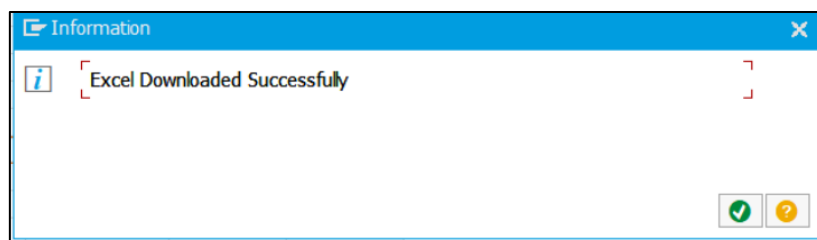
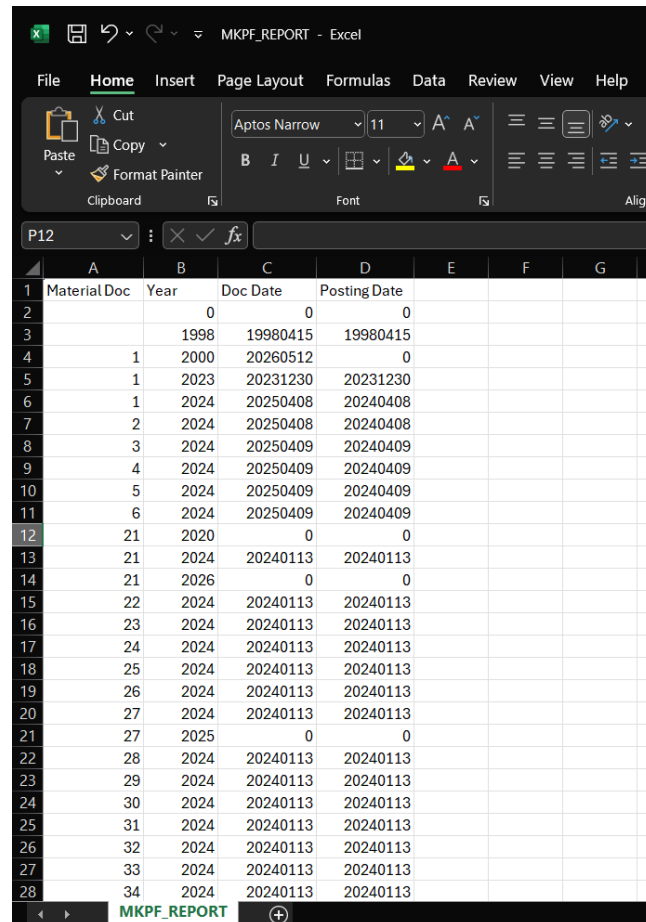


Figure 4 & 5 : Drill-Down ALV Report and Excel Download Functionality

This figure demonstrates the secondary Interactive ALV report generated through drill-down functionality along with the successful Excel export process. When a user selects a material document from the primary ALV report, the system displays detailed item-level information such as Material Number, Quantity, and Unit of Measure. The report can then be exported to Microsoft Excel for further analysis, and the system displays a confirmation message indicating successful completion of the download process.

alv report						
Material Doc	Year	Item	Material	Quantity	UOM	
0000000052	2024	1	000000000000...	10,000	EA	
0000000052	2024	2	000000000000...	1,000	EA	
0000000052	2024	3	000000000000...	1,000	EA	
0000000052	2024	4	000000000000...	1,000	EA	



	A	B	C	D	E	F	G
	Material Doc	Year	Doc Date	Posting Date			
2		0	0	0			
3		1998	19980415	19980415			
4	1	2000	20260512	0			
5	1	2023	20231230	20231230			
6	1	2024	20250408	20240408			
7	2	2024	20250408	20240408			
8	3	2024	20250409	20240409			
9	4	2024	20250409	20240409			
10	5	2024	20250409	20240409			
11	6	2024	20250409	20240409			
12	21	2020	0	0			
13	21	2024	20240113	20240113			
14	21	2026	0	0			
15	22	2024	20240113	20240113			
16	23	2024	20240113	20240113			
17	24	2024	20240113	20240113			
18	25	2024	20240113	20240113			
19	26	2024	20240113	20240113			
20	27	2024	20240113	20240113			
21	27	2025	0	0			
22	28	2024	20240113	20240113			
23	29	2024	20240113	20240113			
24	30	2024	20240113	20240113			
25	31	2024	20240113	20240113			
26	32	2024	20240113	20240113			
27	33	2024	20240113	20240113			
28	34	2024	20240113	20240113			

Figure 5 : Exported Excel Report

This figure shows the inventory movement data exported from SAP into Microsoft Excel. The report contains key transaction details such as Material Document Number, Fiscal Year, Document Date, and Posting Date.

Result:

The exported report enables users to analyze inventory data, prepare business reports, and maintain records outside the SAP system for decision-making and auditing purposes.



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Archive Print and Archive

KREUTZ
21.06.1995
WA

SAP

INVOICE

MATERIAL DOC YEAR	1995
DOCUMENT TYPE	WA
POSTING DATE	21.06.1995
USER NAME	KREUTZ
CHANGED ON	21.06.1995

MAT DOC	MOVE TYPE	PLANT	QTY	AMOUNT
49001008	131	1200	10,000	2.828,11
49001008	261	1200	10.000,000	51,13
49001008	261	1200	11.000,000	1.124,84
49001008	261	1200	11.000,000	281,21
49001008	261	1200	660,000	6,75
49001008	261	1200	100,000	25,56
49001008	261	1200	11,000	50,62
49001008	261	1200	11.000,000	56,24
49001008	261	1200	11.000,000	56,24
49001008	261	1200	15,400	78,74
49001008	261	1200	484,000	247,47
49001008	261	1200	22.000,000	224,97
49001008	261	1200	11.000,000	56,24

TOTAL AMOUNT : 5088.12
IN WORDS : FIVE THOUSAND EIGHTY-EIGHT AND 12 PAISE

SIGNATURE

Figure 6 : Shows the SmartForm-generated Material Document Report displaying header information from the **MKPF** table and item details from the **MSEG** table, including total amount and amount in words in a professional printable format.

Description :

Figure 1 illustrates the SmartForm developed for generating a **Material Document Report** using the **MKPF** and **MSEG** tables. The report displays material document header information such as **Material Document Year, Document Type, Posting Date, User Name, and Changed On Date** along with item details including **Material Document Number, Movement Type, Plant, Quantity, and Amount**. It also calculates the **Total Amount** of all material items and converts the amount into **words** for better readability and business documentation. The report includes the company logo and a professional layout, making it suitable for printing and official inventory records.



The report consists of two main sections:

- **Header Section:** Displays the company logo and inventory document information, including the **Material Document Year, Document Type, Posting Date, User Name, and Changed On Date**. These details are retrieved from the **MKPF (Material Document Header)** table and provide essential information about the inventory transaction.
- **Item Details Section:** Displays a tabular list of all material movement records retrieved from the **MSEG (Material Document Item)** table. Each row contains the **Material Document Number, Movement Type, Plant, Quantity, and Amount**, enabling users to review individual inventory transactions.

At the bottom of the report, the **Total Amount** is calculated by summing the values of all inventory items. The report also converts the total amount into **words**, improving readability and supporting business documentation requirements.

This SmartForm provides a clear and organized representation of inventory movement data and can be used by warehouse personnel, inventory managers, and retail organizations for stock monitoring, inventory valuation, material movement tracking, auditing, and printing inventory transaction reports.



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Results Achieved:

- Improved visibility of inventory movement transactions.
- Faster report generation and data retrieval.
- Reduced manual effort in inventory reporting.
- Enhanced reporting accuracy and operational efficiency.
- Better support for inventory monitoring and supply chain management.

Project GitHub Link

GitHub Repository:

(<https://github.com/sure-trust/AVISHEK-KUMAR-CHOURASIYA-g4-sap-abap>)



New Learnings (Technology, Management, and Professional Skills)

During the SAP ABAP internship, I gained hands-on experience in developing reports using SAP ABAP. I learned how to work with SAP database tables such as MKPF and MSEG, retrieve data using Open SQL, and process information using internal tables and work areas. I developed Interactive ALV reports with drill-down functionality and gained practical knowledge of Smart Forms for generating professional reports. I also learned how to implement Excel export functionality and improve report usability through user-friendly interfaces.

Apart from technical skills, I improved my debugging and problem-solving abilities while resolving development issues during the project. I learned the importance of project documentation, requirement analysis, and systematic testing. Through Soft Skills Training sessions, I enhanced my communication, teamwork, presentation, and time-management skills.

Overall Experience and Reflection

My overall experience during the SAP ABAP internship was highly valuable and enriching. Working on real-time projects helped me understand how SAP solutions are used in business environments to manage inventory and reporting processes. The internship provided practical exposure to SAP development tools and strengthened my understanding of enterprise application development.

Developing the Interactive ALV report, implementing Smart Forms, and integrating Excel export functionality improved my confidence in SAP ABAP programming. .



Conclusion and Future Scope

Recap of Objectives and Achievements

The project aimed to develop a SAP-based solution for tracking inventory movements and generating reports to support retail supply chain operations. The objectives included retrieving inventory data from MKPF and MSEG tables, developing Interactive ALV reports, implementing drill-down functionality, generating Smart Form reports, and enabling Excel export for business analysis.

The project was successfully completed and achieved all planned objectives. An Interactive ALV report was developed to display inventory movement information, drill-down functionality was implemented to provide detailed transaction-level data, Smart Forms were used to generate printable reports, and Excel export functionality was integrated for offline analysis and reporting. The project also demonstrated how SAP solutions can improve inventory visibility, reporting accuracy, and operational efficiency within retail organizations.

Conclusion

The **Retail Inventory Movement Tracking and Reporting System using SAP** successfully demonstrated the role of SAP in improving inventory management and supply chain visibility within the retail sector. By utilizing SAP ABAP technologies and enterprise reporting tools, the project provided an efficient and user-friendly solution for monitoring inventory transactions and generating business reports. The implementation highlighted the importance of automated reporting systems in supporting business operations and decision-making.

The project also provided valuable hands-on experience in SAP ABAP development, Interactive ALV reporting, Smart Forms, database management, and business process understanding. Overall, the project served as a practical demonstration of how SAP



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contributes to operational excellence and digital transformation in modern retail organizations.

Future Scope

Future enhancements may include:

- Integration with SAP MM, SD, and Warehouse Management modules.
- Development of SAP Fiori-based user interfaces.
- Real-time dashboards and graphical analytics.
- Automated email notifications and report scheduling.
- Advanced inventory forecasting and demand analysis.
- Mobile accessibility for inventory monitoring.
- Barcode and QR code-based inventory tracking.
- Integration with SAP Business Intelligence tools.